

Roll No : 01

1. Newton's First Law states that an object remains at rest or in uniform motion unless acted upon by an external force. For example, a book on a table remains at rest until pushed.

Newton's Second law states that the force acting on an object is equal to the rate of change of momentum. It is commonly expressed as $F = Ma$.

Newton's Third Law states that for every action, there is an equal and opposite reaction. For example, when a gun fires a bullet forward, the gun recoils backward.

2. Ohm's law states that the current through a conductor is directly proportional to the voltage across it, provided temperature remains constant.

$$V = IR$$

$$V = 20V$$

$$R = 10\Omega$$

$$I = \frac{V}{R} = \frac{20}{10} = 2A$$

$$\Rightarrow I = 2A$$